

Working with JSON & SQL

Module 16 Study Notes • Advanced Data Ingestion

1. Working with JSON Data

JSON (JavaScript Object Notation) is a universal, text-based data format. Because it represents data as key-value pairs (exactly like Python dictionaries), it is the global standard for transferring data across the internet, particularly via APIs.

Loading JSON into Pandas

- **Local Files:** Use `pd.read_json('data.json')` to instantly convert a structured JSON file into a tabular DataFrame.
- **Remote URLs:** Pandas is smart enough to fetch JSON directly from the web. If an API endpoint returns JSON data, you can simply pass the URL: `pd.read_json('https://api.example.com/data')`.

2. Working with SQL Databases

In enterprise environments, data rarely sits in flat CSV files. It is securely stored in Relational Databases (like MySQL, PostgreSQL, or SQLite). You must learn how to connect Python to these databases to extract your training data.

Step 1: The Database Connector

Python cannot natively talk to a MySQL database. You must install and import a "bridge" library, such as `mysql-connector-python`.

Step 2: Establishing the Connection

You create a connection object by providing the server's credentials:

```
conn = mysql.connector.connect(  
    host='localhost',  
    user='root',  
    password='your_password',  
    database='world'  
)
```

Step 3: Querying Directly into Pandas

Instead of manually fetching SQL rows and appending them to a list, Pandas provides the powerful `pd.read_sql()` function. You pass it a raw SQL query and the connection object, and it returns a fully formatted DataFrame.

```
df = pd.read_sql("SELECT * FROM city WHERE CountryCode = 'IND'",  
conn)
```

3. The Analytical Advantage of SQL

Why query a database instead of just downloading a massive CSV?

- **Memory Efficiency:** By using SQL `WHERE` clauses in your `read_sql()` function, you filter the data *on the database server* before it reaches your machine. This prevents you from overloading your computer's RAM with irrelevant data.
- **Pre-processing:** You can use SQL `JOIN`, `GROUP BY`, and aggregations to clean and format the data at the source, saving you from having to write complex Pandas preprocessing code later.